



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO MAHALAKSHMI & CO

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

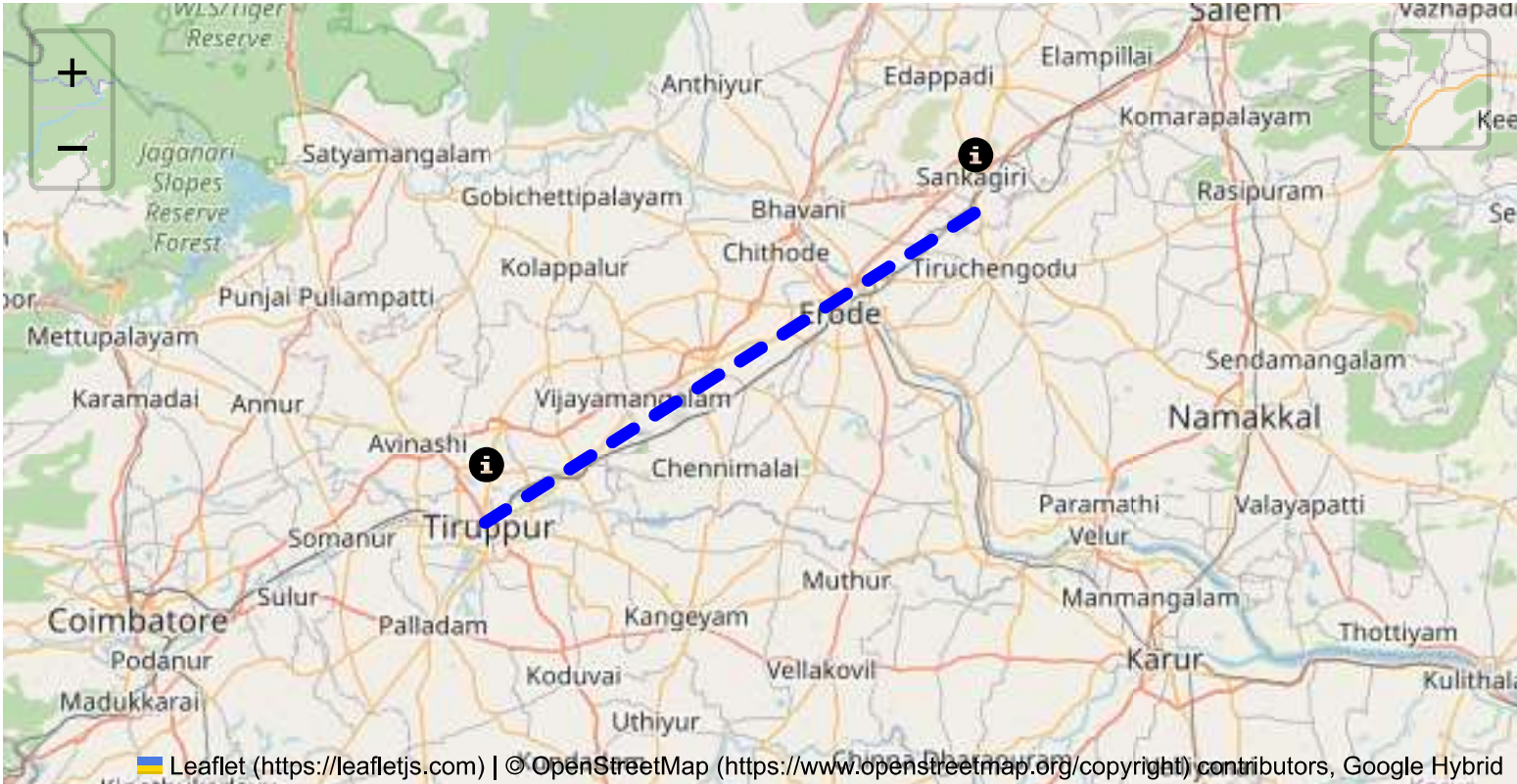
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

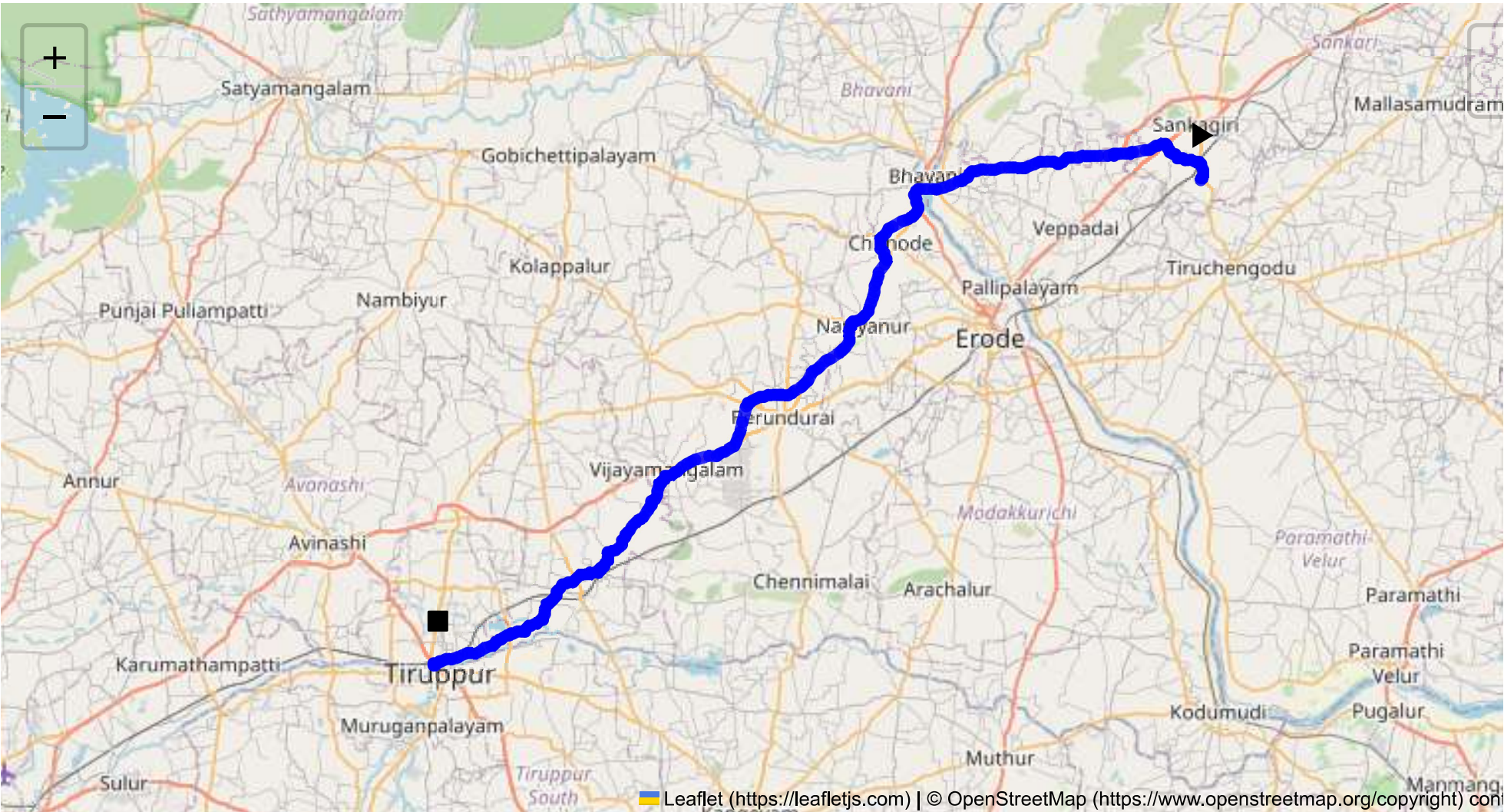
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 81.95 km
Estimated Duration: 1.7 hours
Adjusted Duration (Heavy Vehicle): 2.2 hours
Start: (11.4381, 77.8734)
End: (11.106556, 77.342272)



Welcome to the Journey Risk Management Study

Route Safety Analysis

1. Overview of the Route Map: The route from Sangagiri to Tiruppur spans approximately 81.95 kilometers. Starting from CVQF+23W near Sangagiri, it proceeds via National Highway 544 (NH 544), which is a key arterial road in Tamil Nadu, before turning onto state highways closer to Tiruppur. This route provides a direct and relatively straightforward path between the two points.

2. Typical Weather Conditions and Hazards: Tamil Nadu experiences a tropical climate with high temperatures and humidity levels. The region's weather is characterized by three major seasons: summer (March to June), monsoons (July to September), and winter (October to February). The monsoon season can bring heavy rains, leading to potential flooding and reduced road visibility. Fog can also form in the early mornings during winter, particularly in rural and forested areas, posing additional hazards.

3. Traffic Patterns and Congestion Areas: Peak traffic hours generally occur between 8:30-10:00 AM and 5:00-7:00 PM. NH 544 is likely to experience heavy commercial vehicle traffic due to its status as a major transport route. Congestion is especially prevalent around urban centers like Sankari, Erode, and Perundurai, where local traffic intersects with through traffic.

4. Road Quality and Infrastructure: NH 544 is a well-maintained four-lane highway. However, sections of the road closer to Tiruppur may not be as well-maintained, with potential for potholes and uneven surfaces. Road signage and lighting are generally good on the highway but may be less consistent on state and district roads.

5. Alternative Routes for Emergencies: In case of emergencies, an alternative route could involve taking NH 44 towards Salem and then proceeding via state highways to rejoin the primary route closer to Tiruppur. However, these routes could significantly increase travel time.

6. Local Regulations on Hazardous Material Transport: Tamil Nadu has regulations in place for the transportation of hazardous materials, requiring appropriate labeling and safety documentation. Restrictions may apply to certain areas, especially during peak traffic conditions or special event days.

7. Historical Incidents: The region has seen incidents involving heavy vehicles, often resulting from poor road conditions or driver fatigue. High-profile accidents on NH 544 highlight the importance of vigilance.

8. Environmental Considerations: This route passes near several environmentally sensitive areas, including agricultural lands and minor water bodies. Care should be taken to avoid spills, and appropriate emergency plans should be in place.

9. Communication Coverage: Mobile network coverage is typically strong along NH 544, with potential dead zones in rural or hilly areas, especially when transitioning to state highways nearer to Tiruppur.

10. Emergency Response Times: Emergency services are generally efficient in urban areas but may be slower in rural sections. Expect response times of 30-45 minutes in urban or semi-urban areas, increasing to 60 minutes or more in isolated regions.

11. Overall Summary of Risk Assessment: Traveling through this route involves several risks, including potential adverse weather conditions, heavy traffic, and variable road quality. The transportation of hazardous materials heightens the need for strict adherence to safety protocols. Pre-driving checks, adequate rest for drivers, and constant vigilance are crucial. With proper planning and heightened awareness, these risks are manageable.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44024, 77.87528	15 KM/Hr
2	Turn	High	11.44898, 77.87410	15 KM/Hr
3	Turn	Medium	11.45357, 77.85789	30 KM/Hr
4	Turn	High	11.45352, 77.85698	15 KM/Hr
5	Turn	Medium	11.45847, 77.85140	30 KM/Hr
6	Turn	Medium	11.45898, 77.85135	30 KM/Hr
7	Turn	High	11.46303, 77.84945	15 KM/Hr
8	Turn	Medium	11.45509, 77.81383	30 KM/Hr
9	Turn	Medium	11.45085, 77.77020	30 KM/Hr
10	Turn	Medium	11.43165, 77.67734	30 KM/Hr
11	Turn	Medium	11.23724, 77.50686	30 KM/Hr
12	Turn	High	11.23704, 77.50676	15 KM/Hr
13	Blind Spot	Blind Spot	11.23686, 77.50216	10 KM/Hr
14	Turn	Medium	11.23649, 77.50219	30 KM/Hr
15	Turn	High	11.23396, 77.50105	15 KM/Hr
16	Turn	Medium	11.23329, 77.49966	30 KM/Hr
17	Turn	Medium	11.17851, 77.46169	30 KM/Hr
18	Turn	Medium	11.17368, 77.45730	30 KM/Hr
19	Turn	High	11.16959, 77.44396	15 KM/Hr
20	Turn	Medium	11.16381, 77.43700	30 KM/Hr
21	Turn	Medium	11.16280, 77.43139	30 KM/Hr
22	Turn	Medium	11.15023, 77.42247	30 KM/Hr
23	Turn	Medium	11.13848, 77.41611	30 KM/Hr
24	Turn	Medium	11.13578, 77.41299	30 KM/Hr
25	Turn	Medium	11.12989, 77.40302	30 KM/Hr
26	Turn	Medium	11.12465, 77.38649	30 KM/Hr
27	Turn	Medium	11.12104, 77.38220	30 KM/Hr
28	Blind Spot	Blind Spot	11.10798, 77.34071	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
3	hospital	Dhanvantri Multi Speciality Hospital	11.451362, 77.766602	30 km/h	Medium
4	hospital	Dhanvanthri Hospital	11.4496712, 77.7593772	30 km/h	Medium
5	hospital	J.K.K. Trust Hospital	11.4445841, 77.7307962	30 km/h	Medium
7	hospital	Shri Sathyanarayana Hospital	11.4291297, 77.6913408	30 km/h	Medium
8	hospital	Thanish Siddha Hospital	11.430003, 77.674964	30 km/h	Medium
9	clinic	Harshitha Clinic	11.4313207, 77.674718	30 km/h	Medium
10	hospital	Sri Kaalangi Siddhar Mooligai Vaithiya Nilayam	11.432369, 77.674894	30 km/h	Medium
11	clinic	G.K Clinic	11.4297244, 77.6749715	30 km/h	Medium
12	clinic	Erode Cancer Centre	11.3732, 77.649152	30 km/h	Medium
14	hospital	Irt Hospital	11.2803603, 77.5644118	30 km/h	Medium
16	hospital	Gen Siddha Hospital	11.2375033, 77.5059797	30 km/h	Medium
17	hospital	Vijayamangalm Government Hospital	11.2382827, 77.501687	30 km/h	Medium
19	hospital	Aravind Eye Care Hospital	11.1729469, 77.4574467	30 km/h	Medium
20	clinic	Sri Maha Annai Clinic	11.1707162, 77.4536498	30 km/h	Medium
21	hospital	Ram Doctor Hospital	11.1709391, 77.4532808	30 km/h	Medium
22	clinic	Kumaran Clinic	11.1697553, 77.451543	30 km/h	Medium
23	hospital	Sri Aravindhar Hospital	11.1697756, 77.4448276	30 km/h	Medium
24	clinic	Dr. Raja Clinic	11.1278464, 77.3927768	30 km/h	Medium
25	hospital	Sakthi Sugam Hospital - Tiruppur	11.1151517, 77.3621541	30 km/h	Medium
26	hospital	AG Hospital, Tiruppur	11.1090267, 77.3528045	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
27	hospital	Nallini Hospital	11.1091685, 77.353517	30 km/h	Medium
28	clinic	S R Nursing Home, Tiruppur	11.111432, 77.3489019	30 km/h	Medium
29	clinic	Immuno Lab & Health Care	11.1136966, 77.3487122	30 km/h	Medium
30	police	TIRUPPUR NORTH	11.107892, 77.3450511	30 km/h	Medium
31	hospital	Tirupur Medical Foundation (P) Limited	11.1090003, 77.3449825	30 km/h	Medium
32	police	Tiruppur North	11.1066254, 77.3427939	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
1	college	Vivekanandha Engineering College	11.4589312, 77.7899284	30 km/h	Medium
2	marketplace	Monday market	11.452863, 77.775989	30 km/h	Medium
6	school	SSM Matriculation Higher Secondary School	11.4321653, 77.6880046	30 km/h	Medium
13	college	Government polytechnic college	11.2907053, 77.5698726	30 km/h	Medium
15	school	Bharathi Matriculation School	11.2494613, 77.5307028	30 km/h	Medium
18	college	Sasurie Engineering College	11.2171482, 77.4898552	30 km/h	Medium
33	school	Nanjappa Boys Higher Secondary School	11.1044446, 77.3399245	30 km/h	Medium
34	school	St.Joseph's Matriculation School	11.1046832, 77.3422087	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44024, 77.87528



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45357, 77.85789



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45847, 77.85140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45898, 77.85135



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45509, 77.81383



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45085, 77.77020



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43165, 77.67734



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.23724, 77.50686



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.23704, 77.50676



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.23686, 77.50216



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.23649, 77.50219



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.23396, 77.50105



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.23329, 77.49966



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.17851, 77.46169



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.17368, 77.45730



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.16959, 77.44396



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.16381, 77.43700



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.16280, 77.43139



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.15023, 77.42247



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.13848, 77.41611



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.13578, 77.41299



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.12989, 77.40302



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.12465, 77.38649



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.12104, 77.38220



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.10798, 77.34071
